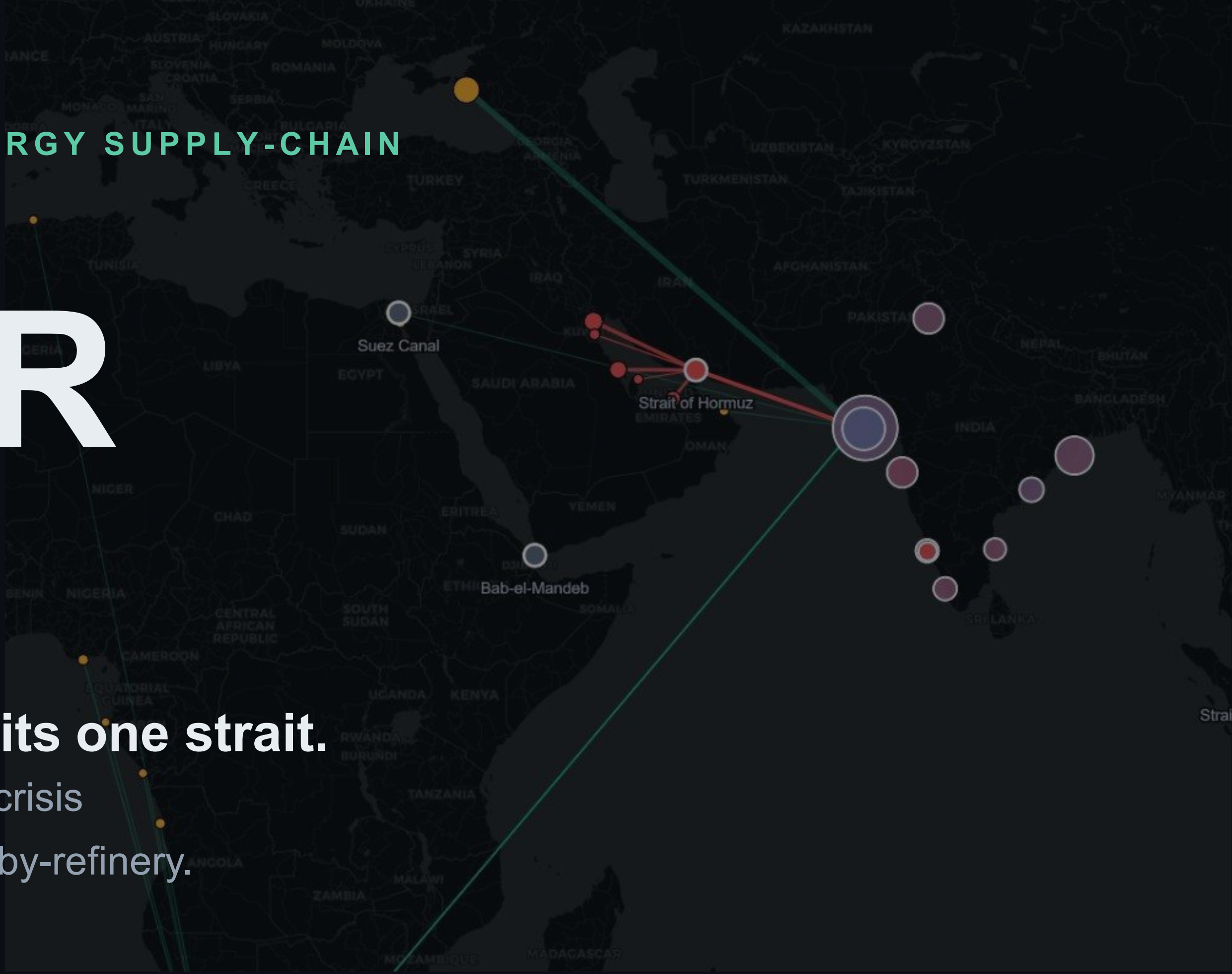
ET AI HACKATHON 2026 · PHASE 2 · PROBLEM #2 — ENERGY SUPPLY-CHAIN RESILIENCE

KARNADHAR

कर्णधार — the helmsman who steers the ship through the storm



India imports 88% of its crude. Nearly half transits one strait.

KARNADHAR is the AI command center that turns a 47-day supply crisis
into a decision made in 45 milliseconds — grade-by-grade, refinery-by-refinery.

Mann Sutariya · Pandit Deendayal Energy University, Gandhinagar

Assumptions reviewed with: Lydia Powell (ORF) · Prof. U. K. Bhui (PDEU)

THE PROBLEM

One strait can strangle the world’s 3rd-largest oil importer

88%
of India’s crude is imported

46%
transits the Strait of Hormuz — derived from official trade records, not assumed

9.5 days
of strategic reserve cover — voluntary, vs the IEA’s 90-day norm

+8%
Brent in a single session when Hormuz was threatened (June 2025)

And the real constraint isn’t barrels — it’s dollars.

India is a twin-deficit economy. A sustained price shock adds ~$190 bn/yr to the import bill, blowing the current-account deficit from 1.2% to over 6% of GDP. Unlike trade-surplus Japan or Korea, India cannot sustain that dollar outflow — which is why unprepared economies take **47 days longer** to stabilise supply (McKinsey). The data to act exists. The intelligence layer to act on it does not.

Sources: DGCIS trade records (May 2025–Apr 2026) · PPAC · ISPRL · EIA · twin-deficit framing reviewed with Lydia Powell, Distinguished Fellow, ORF

THE INSIGHT EVERYONE MISSES

Crude oil is not fungible

Every other “resilience dashboard” treats oil as one number: barrels.

Reality: a refinery is an engine tuned to a fuel. A deep-conversion coking refinery built for heavy, sour, asphaltene-rich crude physically cannot run ultra-light condensate — and a simple refinery cannot digest sour crude. When Hormuz closes, the question is not “where do we buy oil?” It is “which exact grade can each of our refineries actually run — and who has it?”

“That is — you are perfectly right.”

Prof. U. K. Bhui, Petroleum Engineering, PDEU — on our refinery-constrained sourcing thesis. His guidance added the SARA dimension (asphaltenes) to our crude model.

WHAT DECIDES IF A BARREL IS RUNNABLE

- **API gravity** — light vs heavy
- **Sulphur** — sweet vs sour
- **Asphaltene (SARA)** — coke precursor

SAME COUNTRY, DIFFERENT MACHINES

RIL Jamnagar Nelson 21.1
runs almost anything — needs heavy feed

BPCL/HPCL Mumbai Nelson 7.0
light, sweeter diet only — sour is a hard limit

WHAT WE BUILT

A multi-agent brain: signal → scenario → reroute → brief

SIGNAL		SCENARIO		REROUTE		BRIEF
GDELT news agent + live AIS ship tracking	→	glass-box economic cascade + twin deficit	→	grade-aware LP optimizer over real refinery diets	→	executive decision memo, numbers from the engine

- REAL public data — no synthetic signals
- Explicit, testable assumptions (every parameter named & editable)
- Knowledge graph: supplier → route → chokepoint → refinery (50 nodes, 163 edges)
- LLM writes words, never numbers — engine stays auditable

45 ms

end-to-end: signal to executable recommendation

vs the 47-day average for economies without response intelligence

THE FOUNDATION

Built on official trade records — not synthetic data

Refinery (port)	MT/yr	Nelson	Hormuz exposure	Top real suppliers
RIL Jamnagar (Sikka + SEZ)	76.3	21.1	43%	Russia 35 · Saudi 17 · Iraq 14
Nayara Vadinar	41.7	11.8	29%	Russia 56 · Iraq 18 · Angola 6
IOCL Paradip	34.8	12.2	50%	Iraq 24 · Russia 21 · UAE 14
HMEL Bathinda (Mundra)	23.3	12.5	52%	Russia 32 · Saudi 18 · Iraq 16
BPCL/HPCL Mumbai	23.2	7.0	60%	Saudi 25 · UAE 23 · US 21
BPCL Kochi	12.4	8.0	54%	Russia 31 · UAE 23 · Iraq 16

46%

national Hormuz exposure — DERIVED from customs data

10 × 36

refineries × supplier grades, real volumes & landed prices

Every refinery's crude diet, Hormuz dependence, and even its processing limits are grounded in what it actually imported — DGCIS port-wise records, PPAC capacities, public Nelson complexity, crude assay libraries.

DGCIS "India's Import by Principal Commodity" May 2025–Apr 2026 · port→refinery mapping · country→grade assay mapping (crudemonitor/BP/ENI)

THE RESULT THAT MATTERS

The “fungible” plan is physically impossible. Ours is not.

NAIVE · “oil is oil”

INFEASIBLE

250 kb/d of ultra-light condensate assigned to deep coking refineries that **physically cannot run it**. A grade failure — not a logistics failure. This is the plan a fungible model would execute.

KARNADHAR · grade-aware LP

FEASIBLE

Every barrel runnable, sourced, within desulphurisation and asphaltene capacity. Heavy-sour rationed to the cokers (Jamnagar, Paradip); light-sweet routed to simple refineries (Mumbai, Chennai). **Executable by a procurement desk.**

$3.15 M/day

product yield protected by grade-matching

~$1.15 bn/yr

annualised value of that decision quality

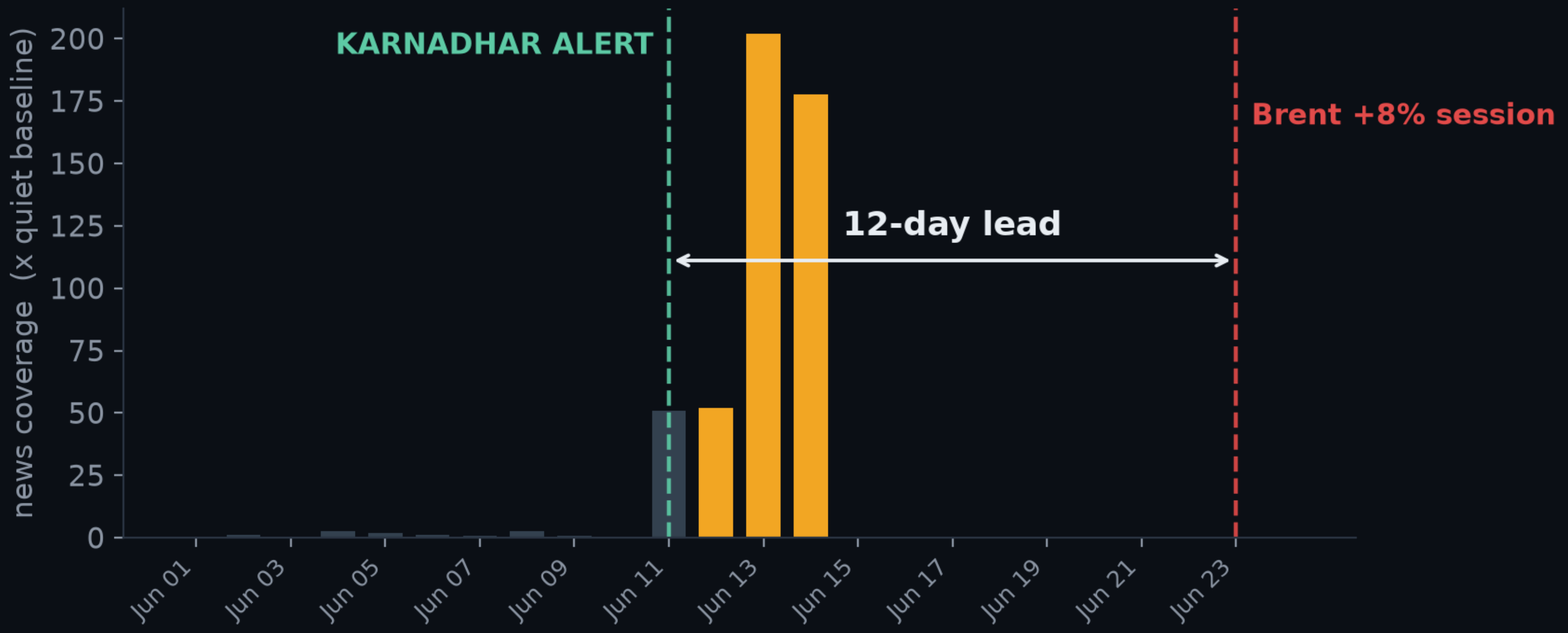
1,370 kb/d

re-sourced across 10 refineries in one solve

Curated stress-demo (15 grades × 7 refineries, public assays) — the wedge REPRODUCES on the real DGCIS model: sulphur-blend breach at Vizag; 220 kb/d un-runnable under sanctions (next slides)

EVALUATION FOCUS 1 · DISRUPTION SIGNAL DETECTION

We saw the June 2025 Hormuz crisis 12 days before the market



12 days

alert lead over the Brent +8% session

200×

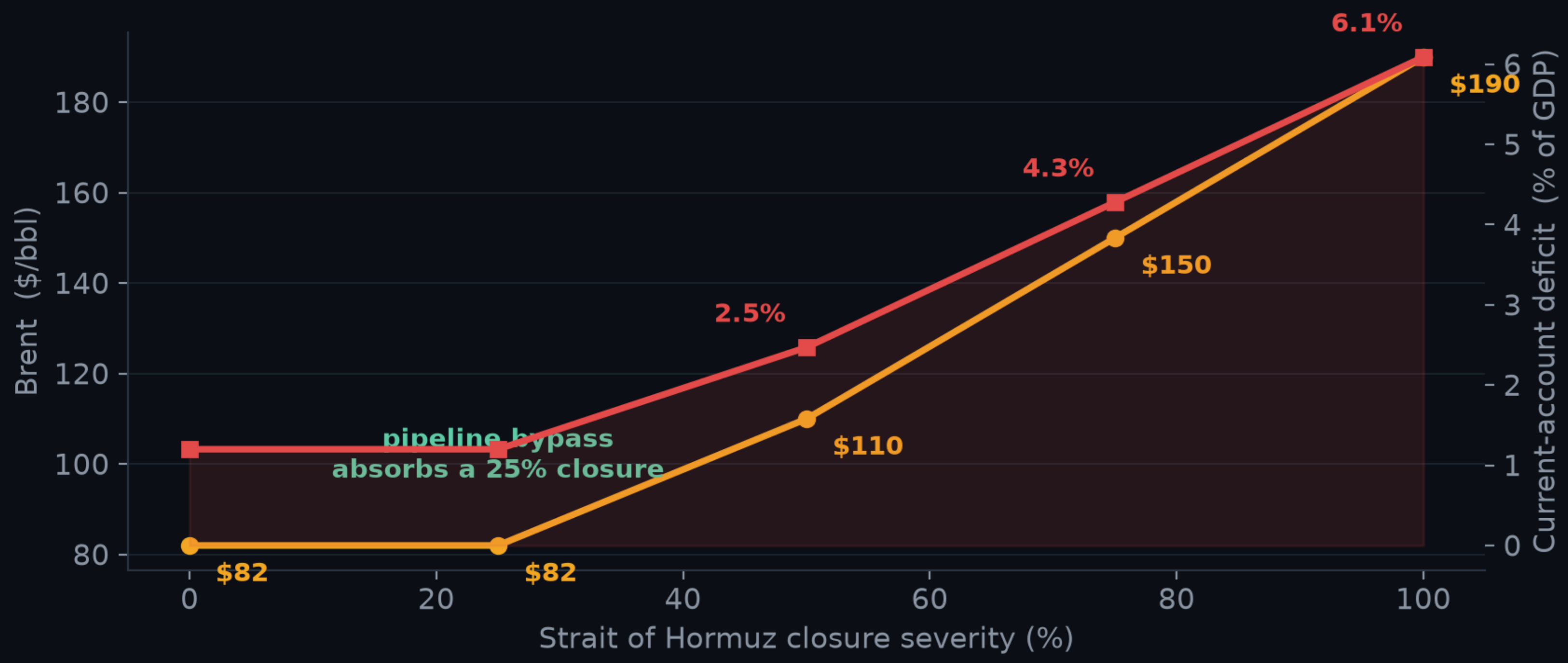
peak news coverage vs quiet baseline (real GDELT)

Method: 10-day rolling baseline → alert when coverage sustains ≥3× baseline for 2 days. Simple, explainable, testable — and fused with live AIS tanker traffic (congestion flags at chokepoints) for multi-source confirmation.

Real GDELT DOC 2.0 coverage-volume series, cached 1–14 June 2025 · backtest vs 23 June Brent session · aisstream.io live vessel feed (Malacca verified live)

EVALUATION FOCUS 3 · SCENARIO MODEL FIDELITY

A glass-box cascade — ending in the deficit India cannot pay



₹125.5/L

retail fuel at full closure (+₹25.5)

−1.4 pp

GDP drag · +2.0 pp CPI

$191 bn/yr

extra import bill in USD

1.2% → 6.1%

current-account deficit, % of GDP

Every parameter is named, sourced and editable; sensitivity is one function call. “The binding constraint is the balance of payments, not the barrels” — the twin-deficit reframe came from our review with ORF.

Cascade: closure → net shortfall (after Saudi/UAE pipeline bypass) → Brent → import bill → pump price → CPI/GDP → CAD · reviewed with Lydia Powell, ORF

BEYOND ONE SCENARIO

Block any chokepoint. Sanction any supplier. Model it.

Hormuz full closure	naive: INFEASIBLE — blend breaches Vizag's sulphur limit	KARNADHAR: FEASIBLE — and priced: the reroute ties up +66 VLCC-equivalents (ton-mile)
Hormuz + OPEC+ squeeze	alternate suppliers squeezed to 60% of headroom	still feasible — grade-matching value rises to $1.95M/day of protected yield
Russia sanctioned	naive sends 220 kb/d to refineries that cannot run it	usable shortfall 220 → 0 — India's #1 supplier is replaceable, at a price
Hormuz + Russia (compound)	the nightmare: 3,900 kb/d gap, 2 of top 3 suppliers gone	honest: 2,166 kb/d unmet, quantified — naive pretends, with 546 kb/d of fake fill

Also modelled: the brief's Red Sea suspension — India's crude barely moves (Cape-routed), but the commodity lens shows who is hit: Black-Sea edible oils (18%). And when a shock is too severe to reroute around, KARNADHAR quantifies the gap — the SPR scheduler answers hold / bridge / ration with a drawdown rate.

EVALUATION FOCUS 4 · GEOSPATIAL INTELLIGENCE

The war-room: India’s energy lifeline, live on one screen

Optimizer flows — the plan, drawn
every LP allocation becomes an animated flow: source → corridor → the exact refinery, width = kb/d

Live 3D globe / 2D toggle
true globe projection (or flat map), real AIS vessels, bypass pipelines mapped

Interactive knowledge graph
zoom, pan, drag the 50-node supplier→strait→refinery web; threat edges glow per scenario

Multi-commodity lens
same disruption scored across 8 strategic imports — Hormuz hits 53% of LNG too

Decision layer
shadow prices, VLCC ton-mile, SPR drawdown schedule + executive brief — on every scenario

WHY YOU CAN TRUST IT

Validated, evaluated, and reviewed by domain experts

AUTOMATED VALIDATION

50 / 50

checks passed — one command, exit-code gated, run in CI on every push

- ✓ wedge holds: naive infeasible for grade reasons, LP optimal
- ✓ cascade monotonic; bypass nuance; sensitivity band; twin deficit
- ✓ signal alert precedes market repricing on real data
- ✓ real-diet integrity: 46% exposure, configs sane
- ✓ engine admits unmet demand; SPR scheduler holds/bridges/rations
- ✓ decision layer proven: shadow prices · VLCC ton-mile · usable shortfall
- ✓ multi-commodity lens generalises — Red Sea spares crude, hits edible oils

EVALUATION FOCUS — ALL FIVE, COVERED

EF-1	signal lead time: 12 days, real GDELT + AIS
EF-2	procurement executability: feasible, grade-matched plans
EF-3	scenario fidelity: explicit, testable, sensitivity-swept
EF-4	geospatial depth: live routed map, real diets
EF-5	response time: 45 ms signal → recommendation

EXPERT REVIEW

Lydia Powell — Distinguished Fellow, ORF: twin-deficit framing; SPR-as-voluntary-insurance; sourcing-first policy sequence.

Prof. U. K. Bhui — Petroleum Engg., PDEU: wedge validated (“you are perfectly right”); SARA/asphaltene dimension added on his guidance.

BUSINESS IMPACT & SCALABILITY

Incumbents tell you what's happening. We tell you what to do.

Kpler, Vortexa, S&P and Wood Mackenzie built a multi-billion-dollar market selling **monitoring**. KARNADHAR sells the **decision layer** — the feasible, grade-aware action plan — AI-native and India-first.

WHO PAYS

Refiners
IOCL · BPCL · HPCL · RIL · Nayara

Traders
desks living on disruption signals

Shippers & insurers
chokepoint risk = pricing

Government
PPAC · ISPRL · MoPNG — the anchor

THREE AXES OF SCALE

1 · Segments refiner → trader → insurer → sovereign (same engine, 7 buyer types)

2 · Geography 50+ import-dependent economies share this exact anatomy — Japan, Korea, Turkey, Philippines…

3 · Commodity demonstrated LIVE in-app: 8 strategic imports screened — LNG, pharma APIs, semiconductors… (Hormuz hits 53% of LNG)

Model: tiered SaaS — Monitor → Simulate → Orchestrate — plus API licensing. Land commercial desks first (months, not years); the sovereign deployment is the moat and the mission.

NEXT 90 DAYS

- → Refinery configs calibrated with Prof. Bhui’s written inputs (in progress)
- → Commercial AIS feed for dense Gulf coverage; Claude-authored live briefs
- → Pilot conversations: PPAC · ISPRL — the analysis they do in days, in 45 ms

India’s energy lifeline has no helmsman for the storm.

Now it does.

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